

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 14



By- Vishal Sir

Recap of Previous Lecture



Topic

Bounded lattice



Topic

Complements of an element in a lattice



Topic

Complemented lattice



Topics to be Covered



Topic

Complements of an element in a lattice



Topic

Sub-lattice



Topic

Distributive lattice



H.W. :- Find the complement of Every element (if exist)
of the following lattices.

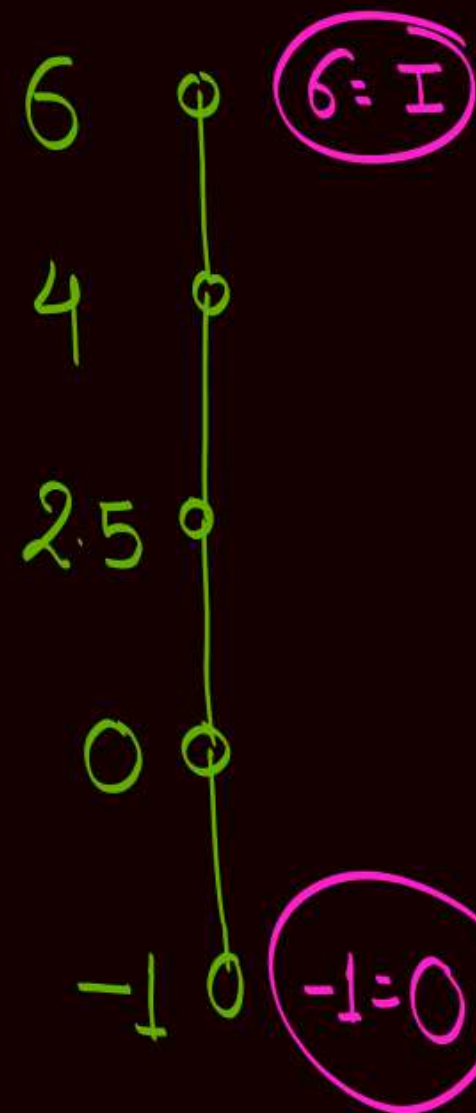
(i) $(\{-1, 0, 2.5, 4, 6\}, \leq)$

$$\begin{array}{l} -1 \vee 6 = 6 = I \quad \therefore \overline{-1} = 6 \\ -1 \wedge 6 = -1 = 0 \quad \therefore \overline{6} = -1 \end{array}$$

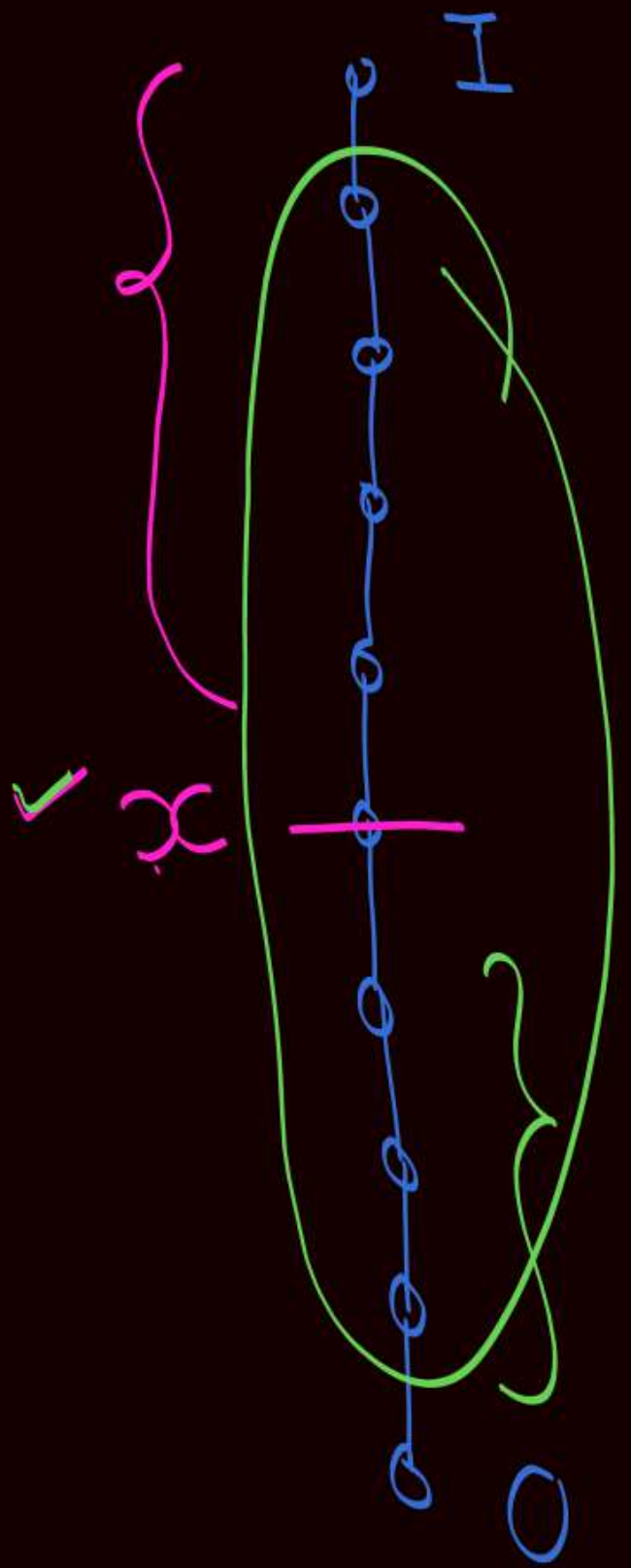
$\overline{0}$ = does not exist

$\overline{2.5}$ = does not exist

$\overline{4}$ = does not exist



There are some elements for which Complement does not exist, $\therefore$ It is not a Complemented lattice



Note:- ① A TOSET is always a lattice, but
Every lattice need not be a TOSET.

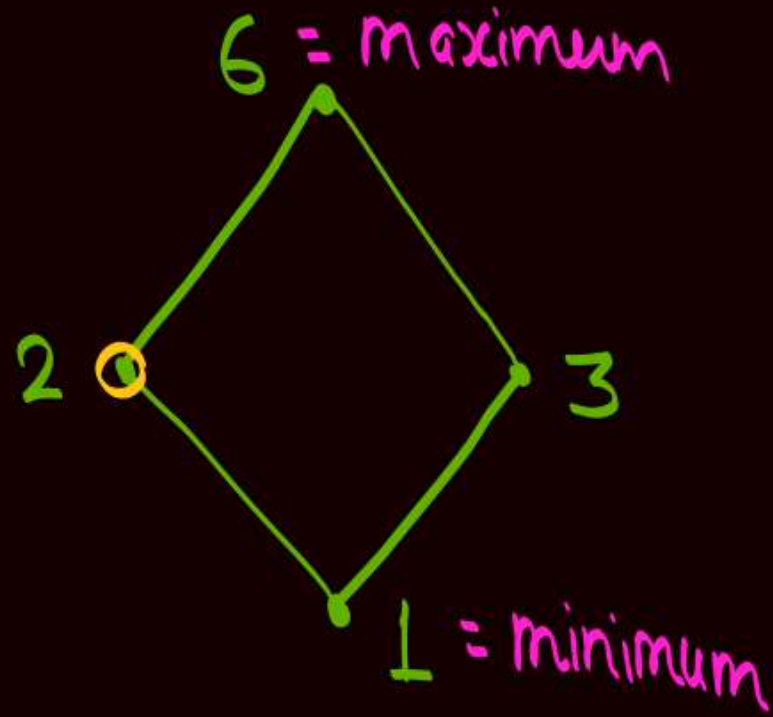
② If TOSET is a bounded lattice, then
Complement of minimum element = maximum element
And Complement of maximum element = minimum element
but Complement does not exist for any other
element in a lattice w.r.t. TOSET

③ If two elements lie on the same line in a
Hasse diagram, then they can not be complement
of each other, Except for minimum & maximum element

H.W. :- Find the complement of Every element (if exist)
of the following lattices.

(ii) $(D_6, \div)$

$$\begin{aligned}\bar{1} &= 6 \\ \bar{6} &= 1\end{aligned}$$



$2 \vee 3 = 6 = \text{maximum}$
 $2 \wedge 3 = 1 = \text{minimum}$ } Hence 2 & 3 are
Complement of each other

∴

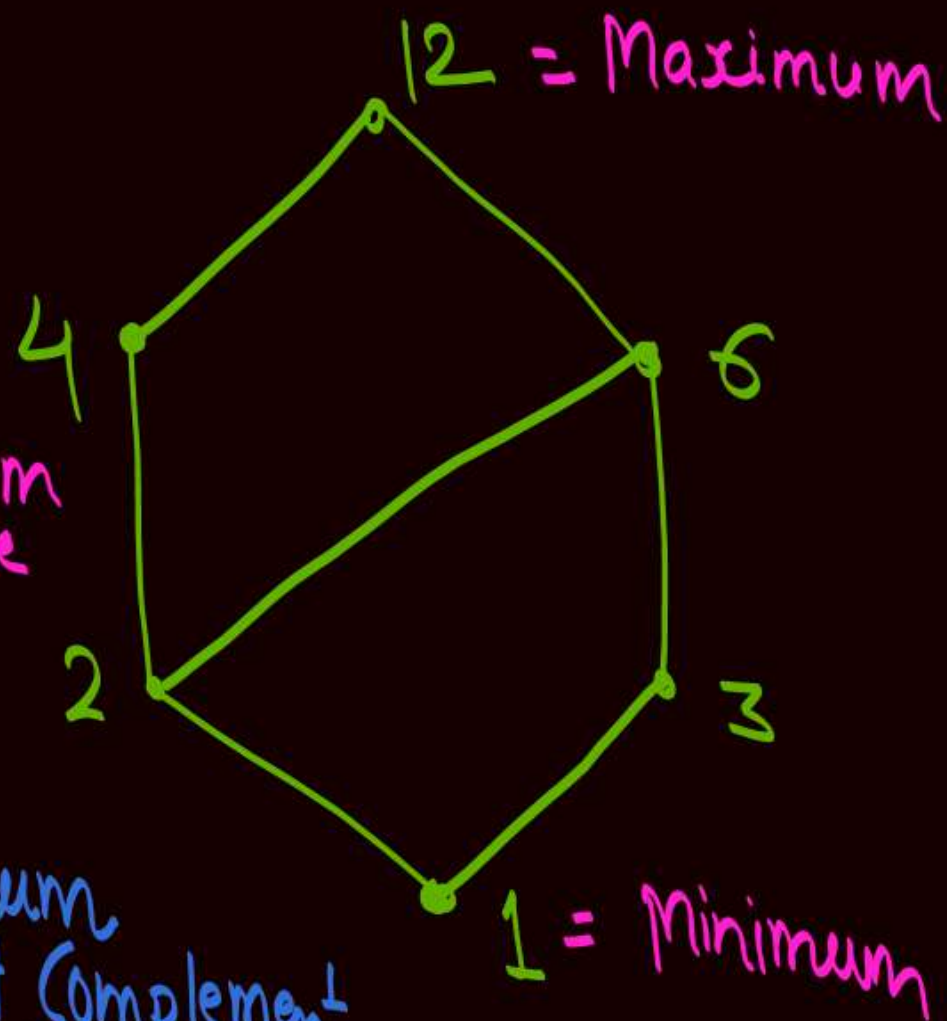
$$\begin{aligned}\bar{2} &= 3 \\ \bar{3} &= 2\end{aligned}$$

$$\begin{aligned}\bar{1} &= 6 \\ \bar{6} &= 1 \\ \bar{2} &= 3 \\ \bar{3} &= 2\end{aligned}$$

Every element of
the lattice has
at least '1' complement
∴ Given lattice is a
Complemented lattice

H.W. :- Find the Complement of Every element (if exist)
of the following lattices.

(iii) $(D_{12}, \div)$



$$\overline{1} = 12$$

$$\overline{2} = \text{does not exist} \Rightarrow 2 \vee 3 = 6 \neq \text{Maximum}$$

$\therefore 2 \nmid 3$ can not be
Complement of
each other

$$\overline{3} = 4$$

$$\overline{4} = 3,$$

$$\overline{6} = \text{does not exist}$$

$$\overline{12} = 1$$

$$\Rightarrow 4 \wedge 6 = 2 \neq \text{Minimum}$$

$\therefore 4 \nmid 6$ are not Complement
of each other

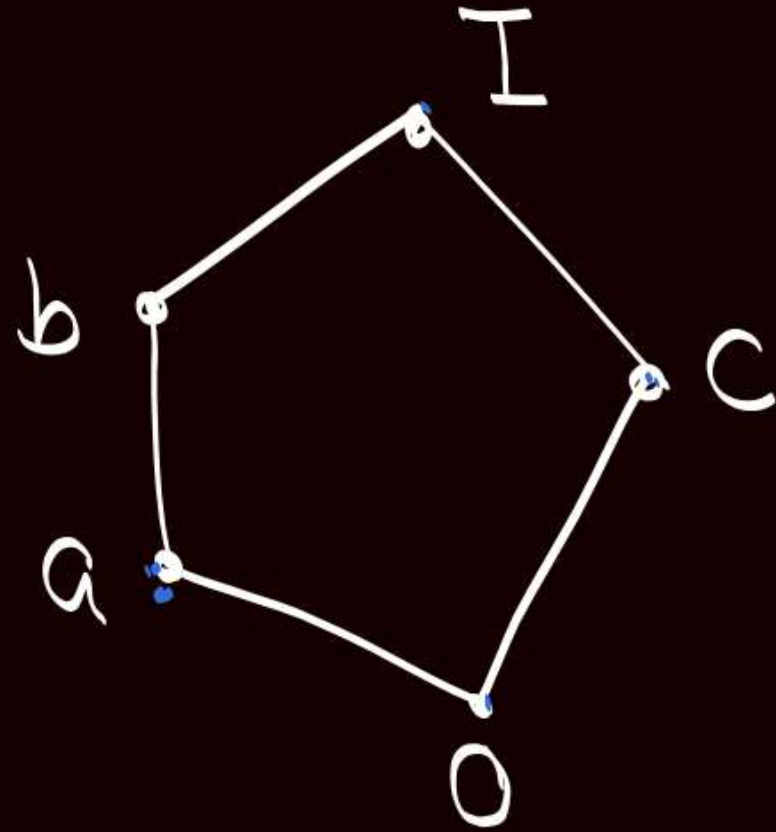
$$3 \vee 4 = 12 = \text{maximum}$$

$$3 \wedge 4 = 1 = \text{Minimum}$$

$\therefore 3 \nmid 4$ are Complement
of each other

Complement does not exist
for some elements of
lattice, $\therefore$ Not a
Complemented lattice

eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



$$\overline{0} = I$$

$$\overline{a} = c$$

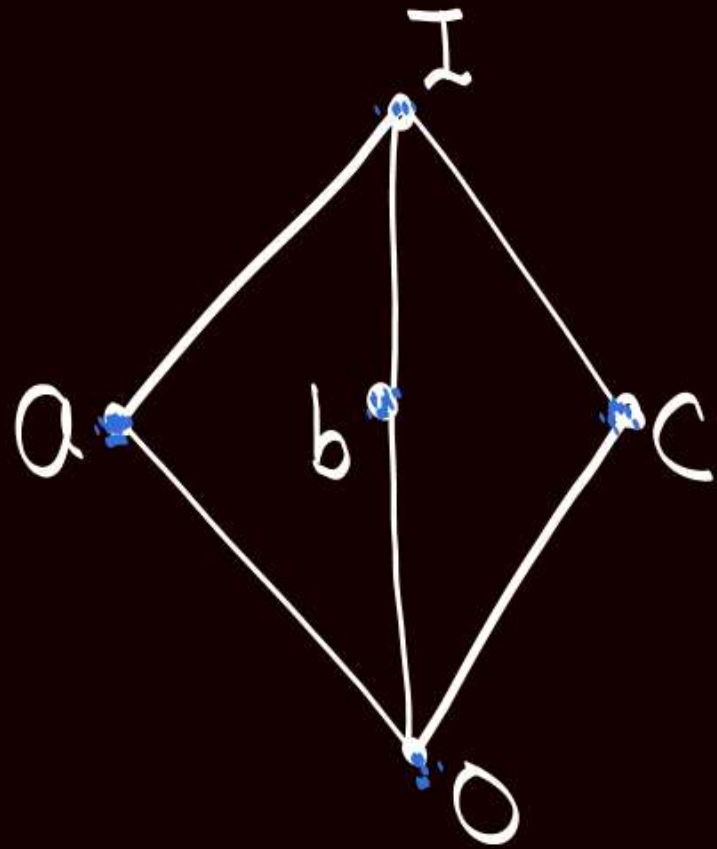
$$\overline{b} = c$$

$$\overline{c} = a, b$$

$$\overline{I} = 0$$

Every element has at least one complement
i.e. given lattice is complemented lattice

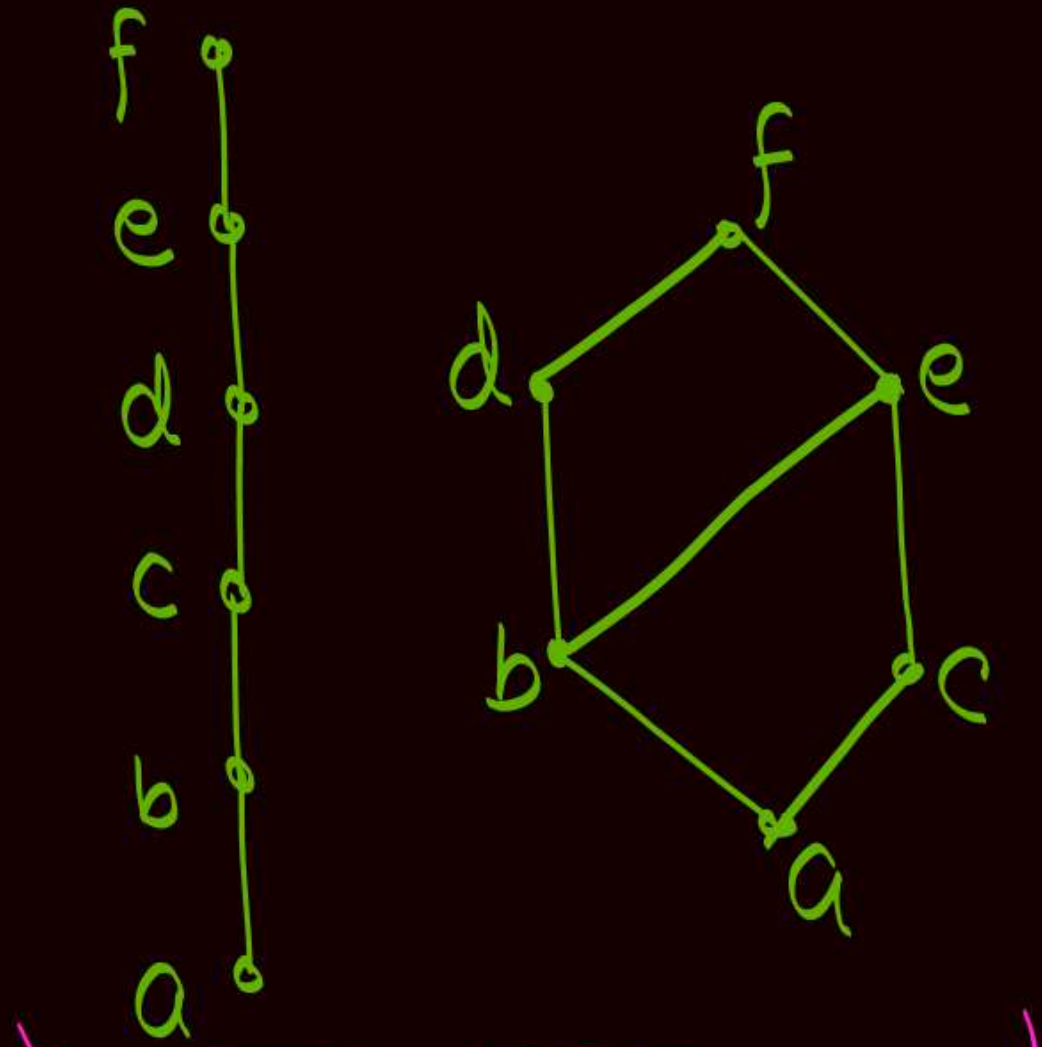
eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



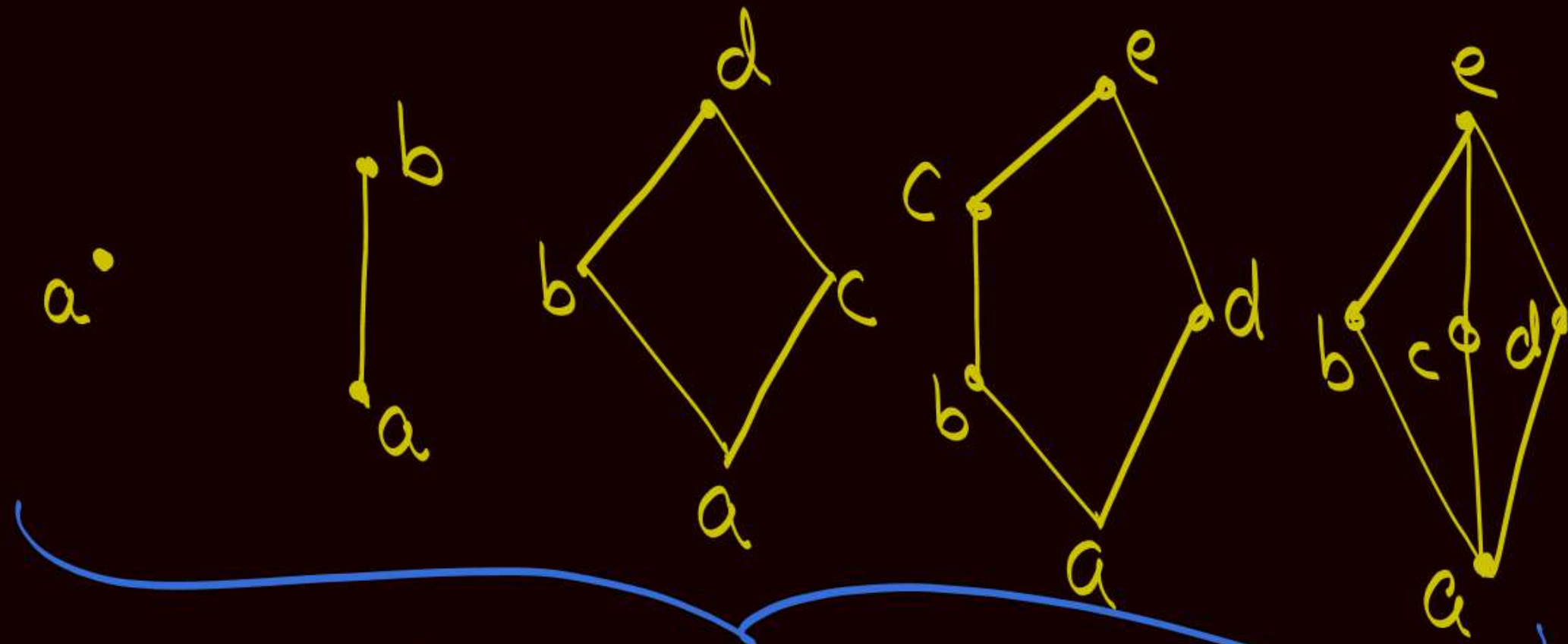
$$\begin{aligned}\overline{0} &= I \\ \overline{a} &= b, c \\ \overline{b} &= a, c \\ \overline{c} &= a, b \\ \overline{I} &= 0\end{aligned}$$

Every element has at least one complement
∴ given lattice is complemented lattice

Note:-



Not Complemented lattice



all are Complemented lattice.



Topic : Sublattice

Let $[L, \vee, \wedge]$ be a lattice, and M is any subset of set L

' M ' is a sublattice of lattice L

if and only if, (i) $[M, \vee, \wedge]$ must be a lattice

{ i.e. lub & glb must exist for every }
Pair of elements of set M

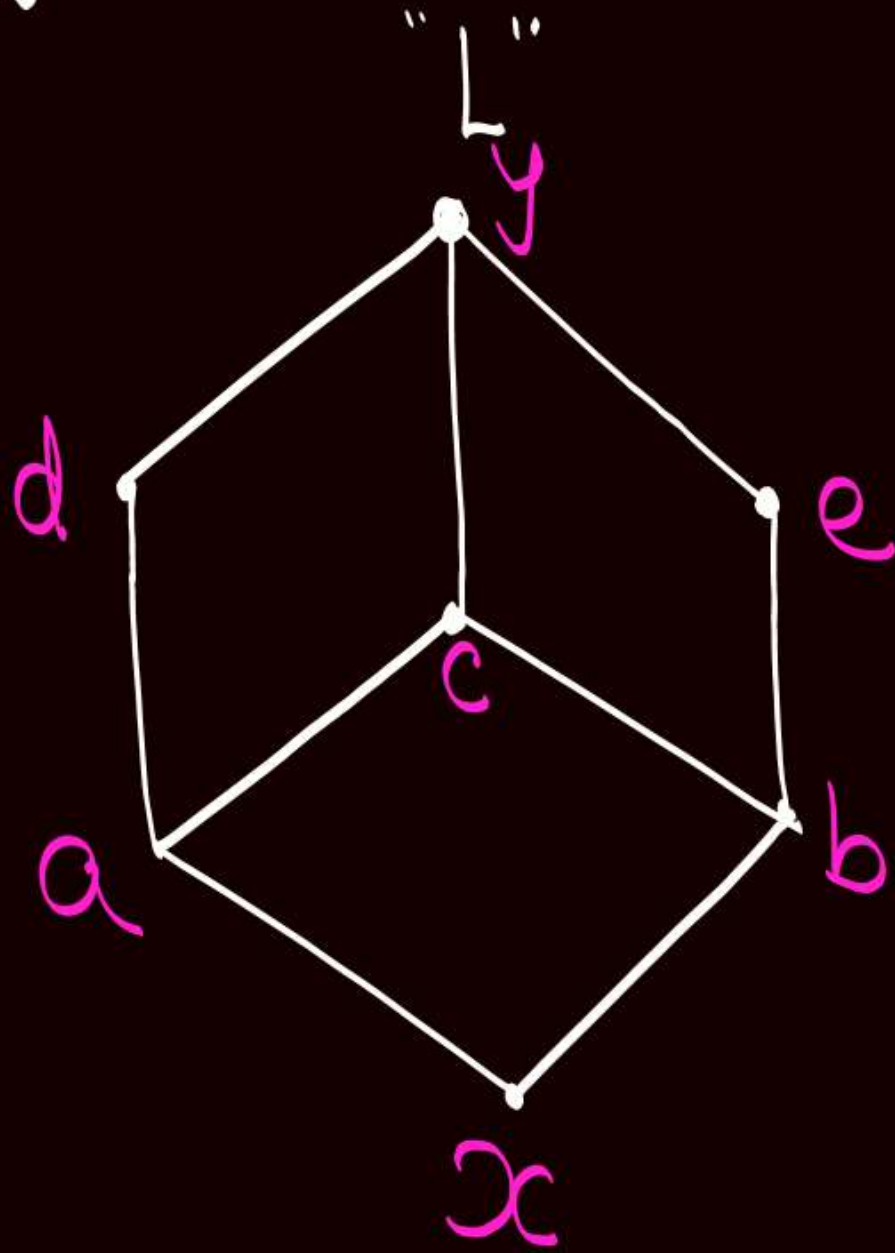
(ii) lub and glb for any pair of elements in lattice $[M, \vee, \wedge]$ must be exactly same as lub and glb for that pair of element in lattice $[L, \vee, \wedge]$

i.e. $\forall a, b \in M$

$$\text{lub}(a, b)_{\text{in } M} = \text{lub}(a, b)_{\text{in } L}$$

$$\& \text{glb}(a, b)_{\text{in } M} = \text{glb}(a, b)_{\text{in } L}$$

eg Consider the following Lattice 'L'



Which of the following is/are sublattices of lattice L

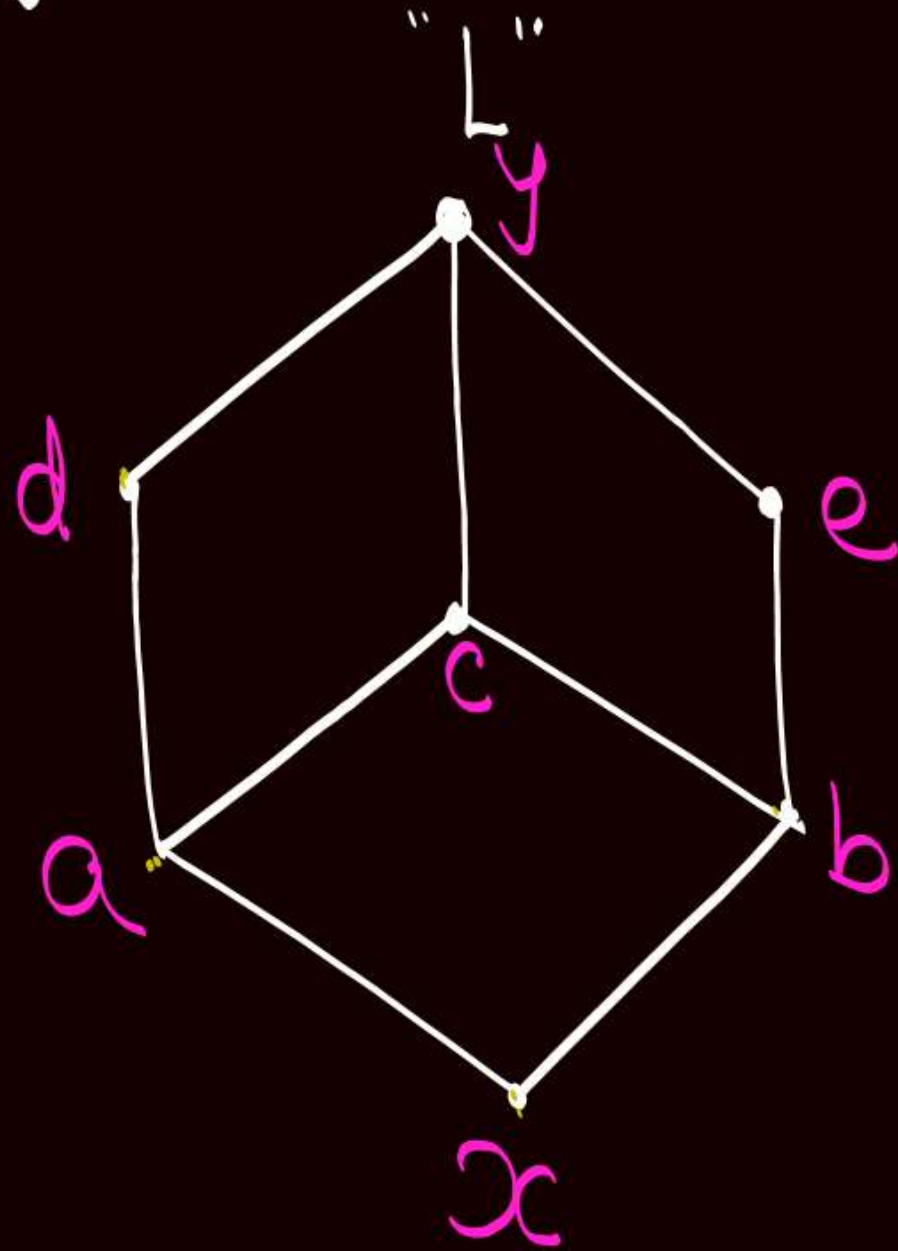
(i) $L_1 = \{x, a, b, y\}$

(ii) $L_2 = \{x, c, d, y\}$

(iii) $L_3 = \{x, a, c, y\}$

(iv) $L_4 = \{x, a, e, y\}$

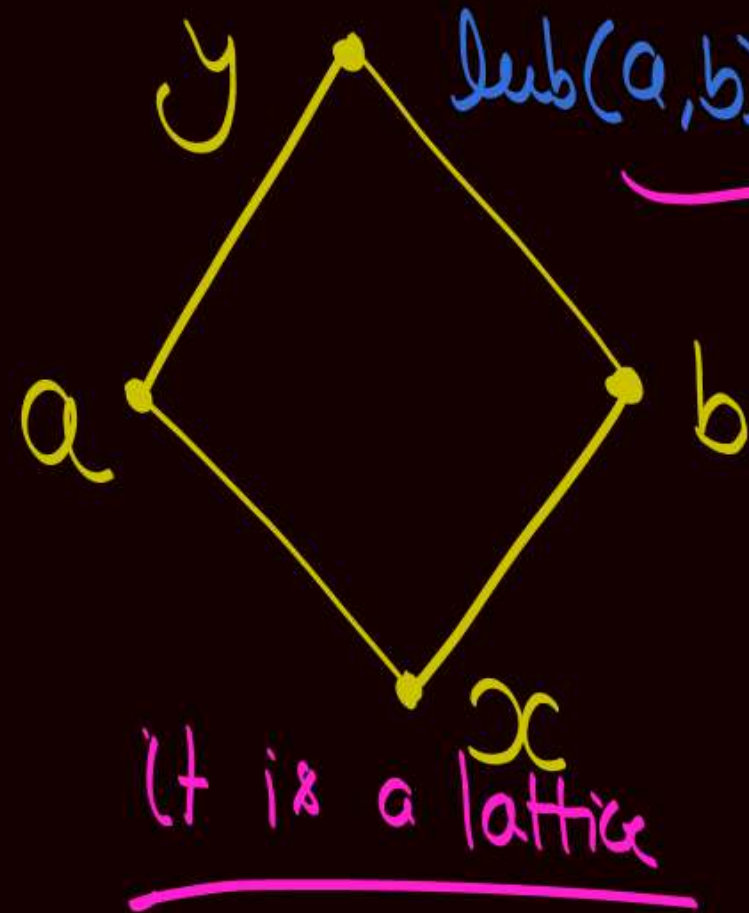
eg Consider the following Lattice 'L'



$$\text{lub}(a, b) = c$$

Which of the following is/are sublattices of lattice L

(i) $L_1 = \{x, a, b, y\}$

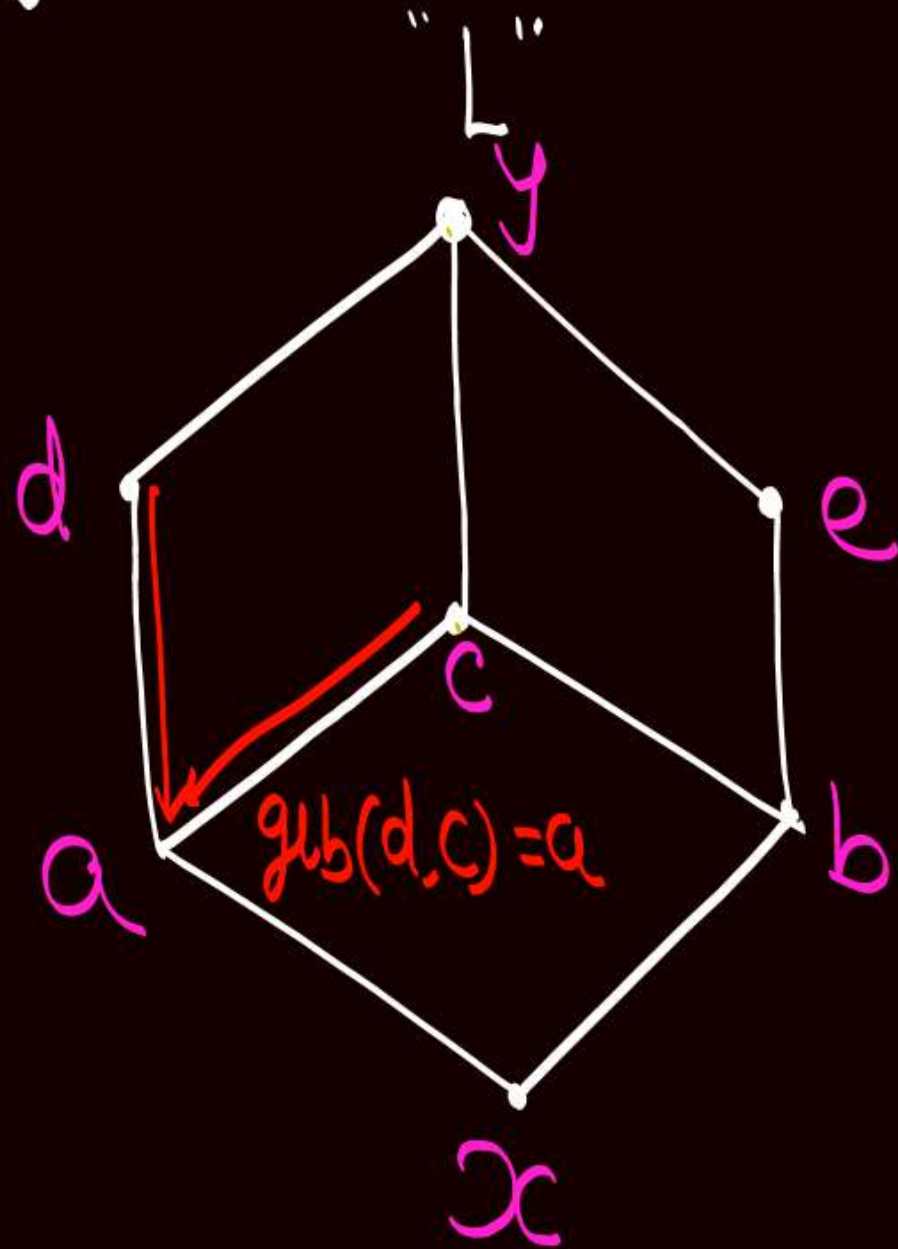


$$\text{lub}(a, b) = y \neq c$$

ie; Not same as L
 $\therefore L_1$ is not a sub-lattice of L

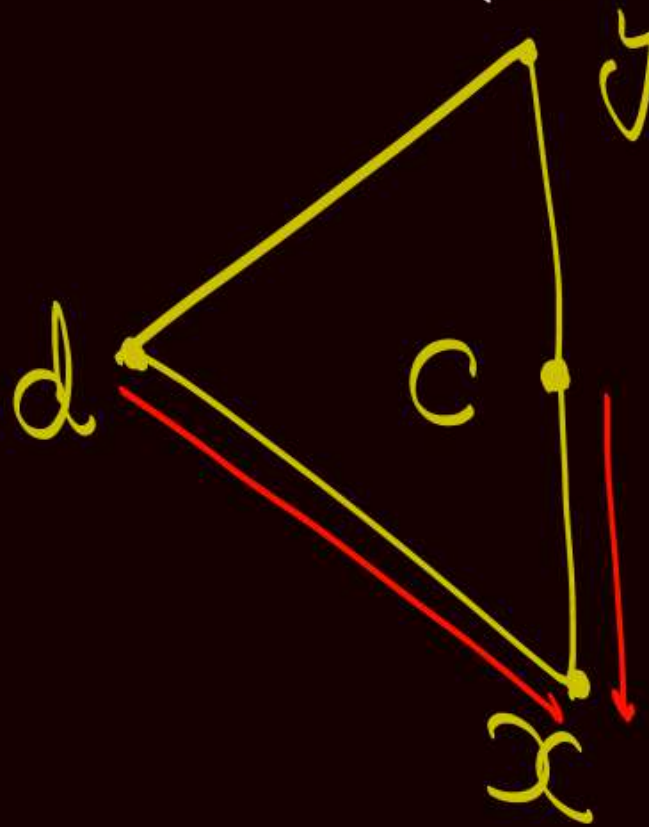
It is a lattice

eg Consider the following lattice 'L'



Which of the following is/are sublattices of lattice L

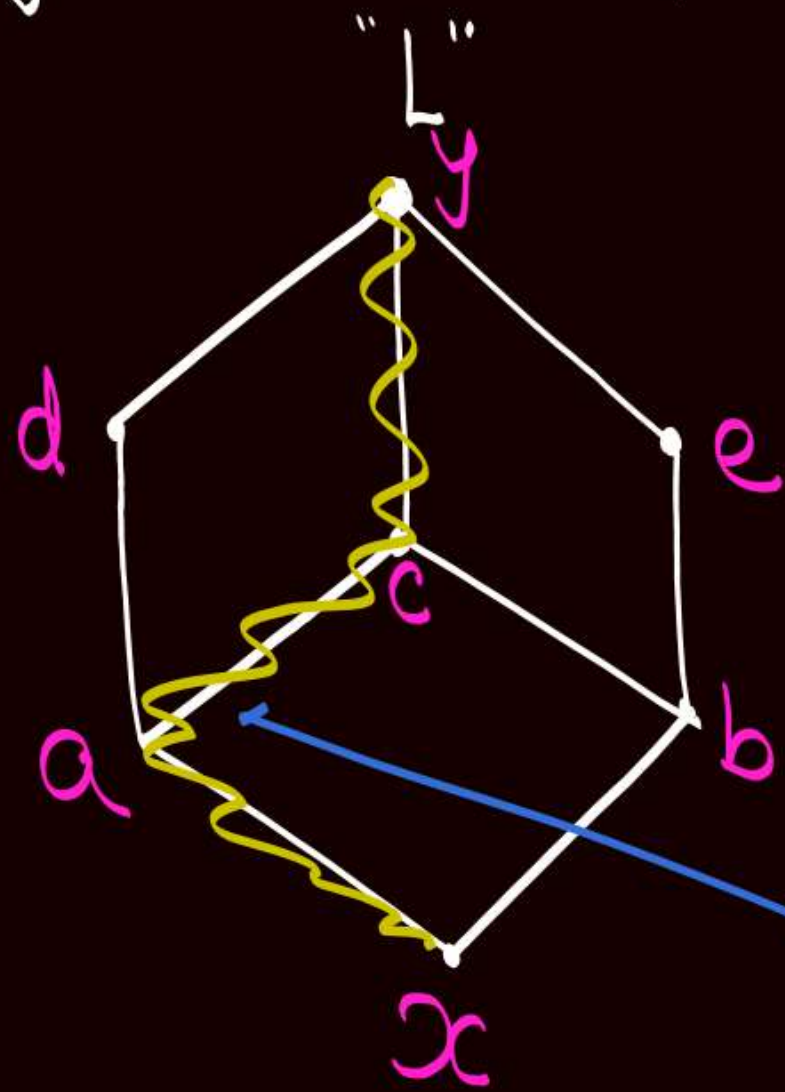
(ii) $L_2 = \{x, c, d, y\}$



$\underbrace{d \vee c = x \neq a}$

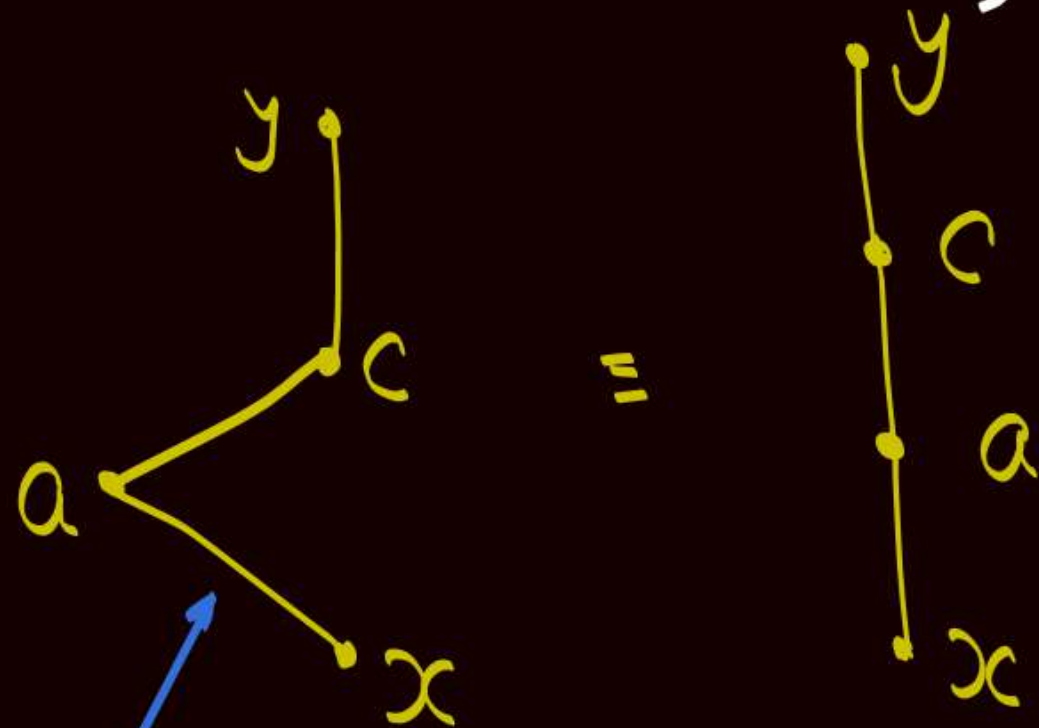
Not same as lattice L
∴ Not a sublattice of L

eg Consider the following Lattice 'L'



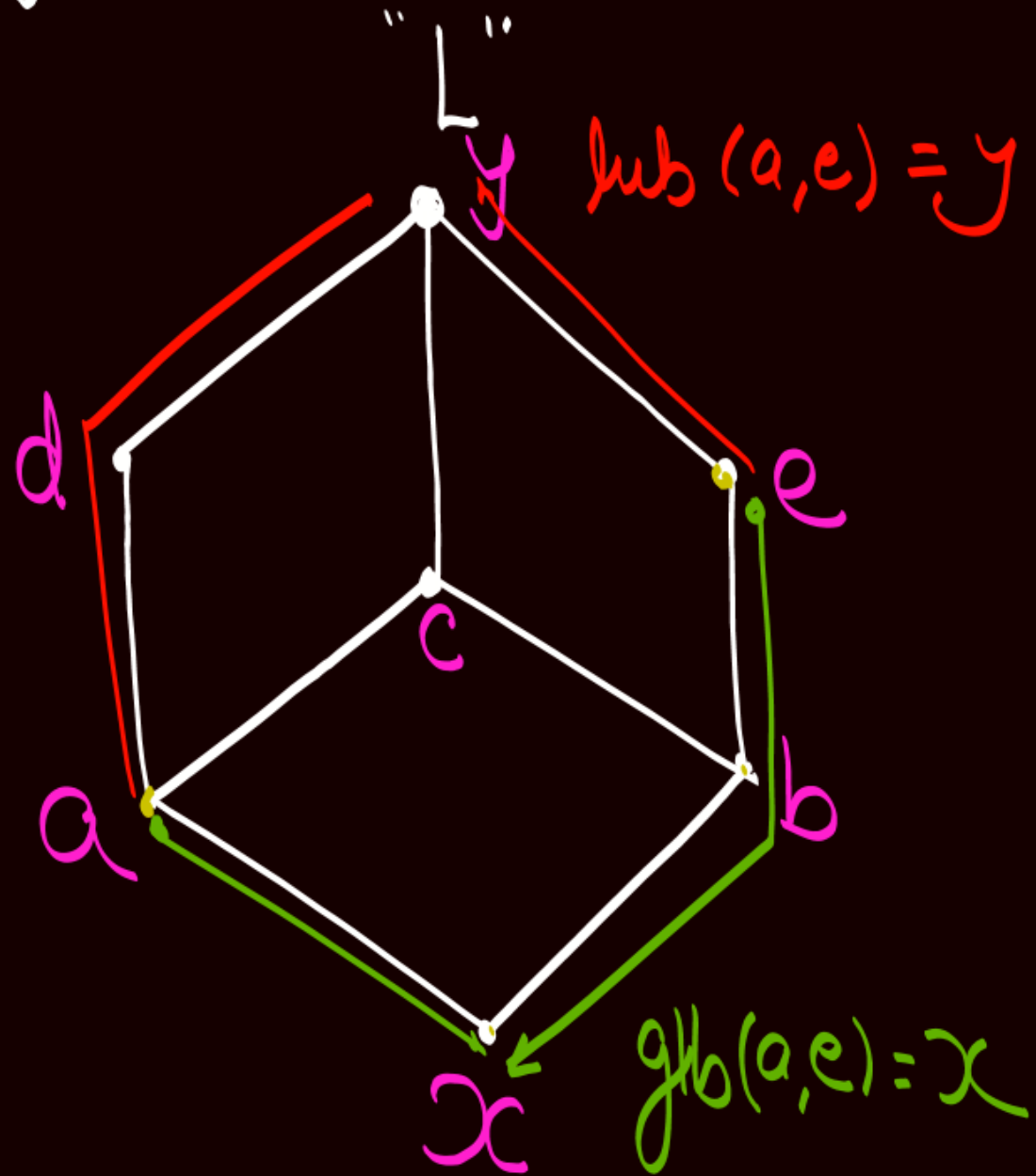
Which of the following is/are sublattices of lattice L

(iii) $L_3 = \{x, a, c, y\}$



Elements are on the same line
 ∴ lub & glb for every pair of elements in L_3 will be exactly same as 'L', ∴ it is a sub-lattice of L.

eg Consider the following Lattice 'L'

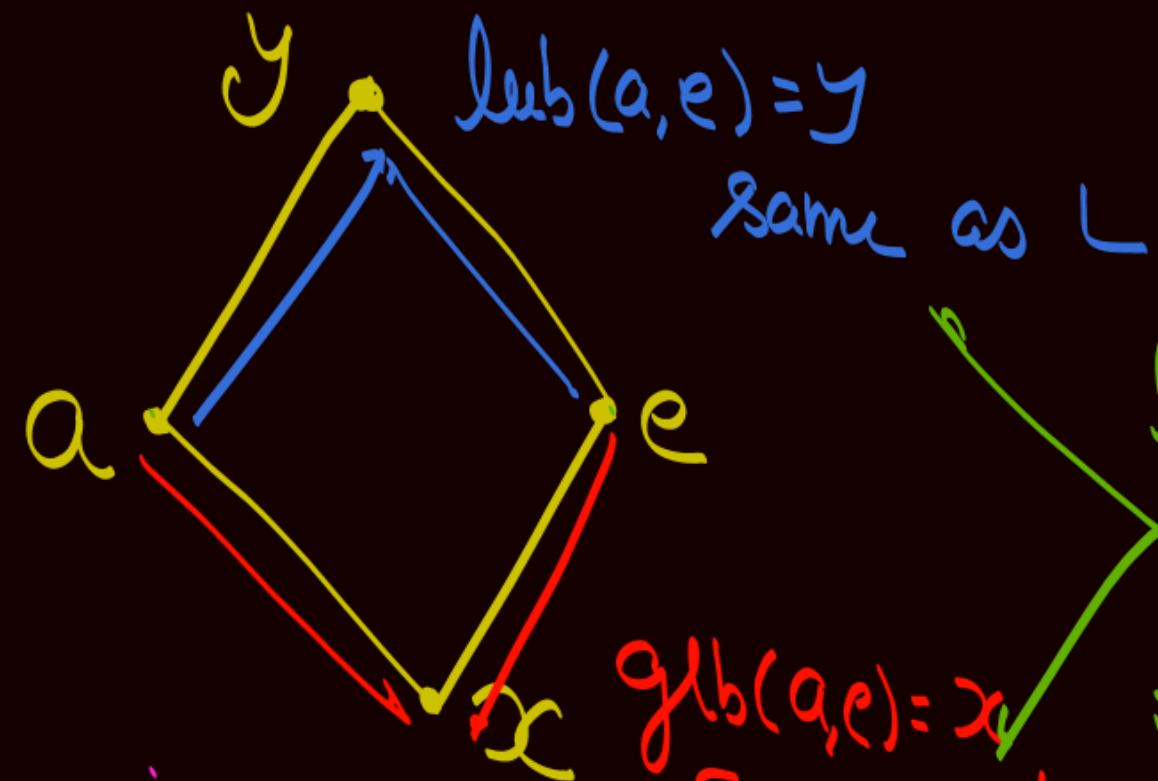


$$\text{lub}(a, e) = y$$

$$\text{glb}(a, e) = x$$

Which of the following is/are sublattices of lattice L

(iv) $L_4 = \{x, a, e, y\}$



$$\text{lub}(a, e) = y$$

Same as L

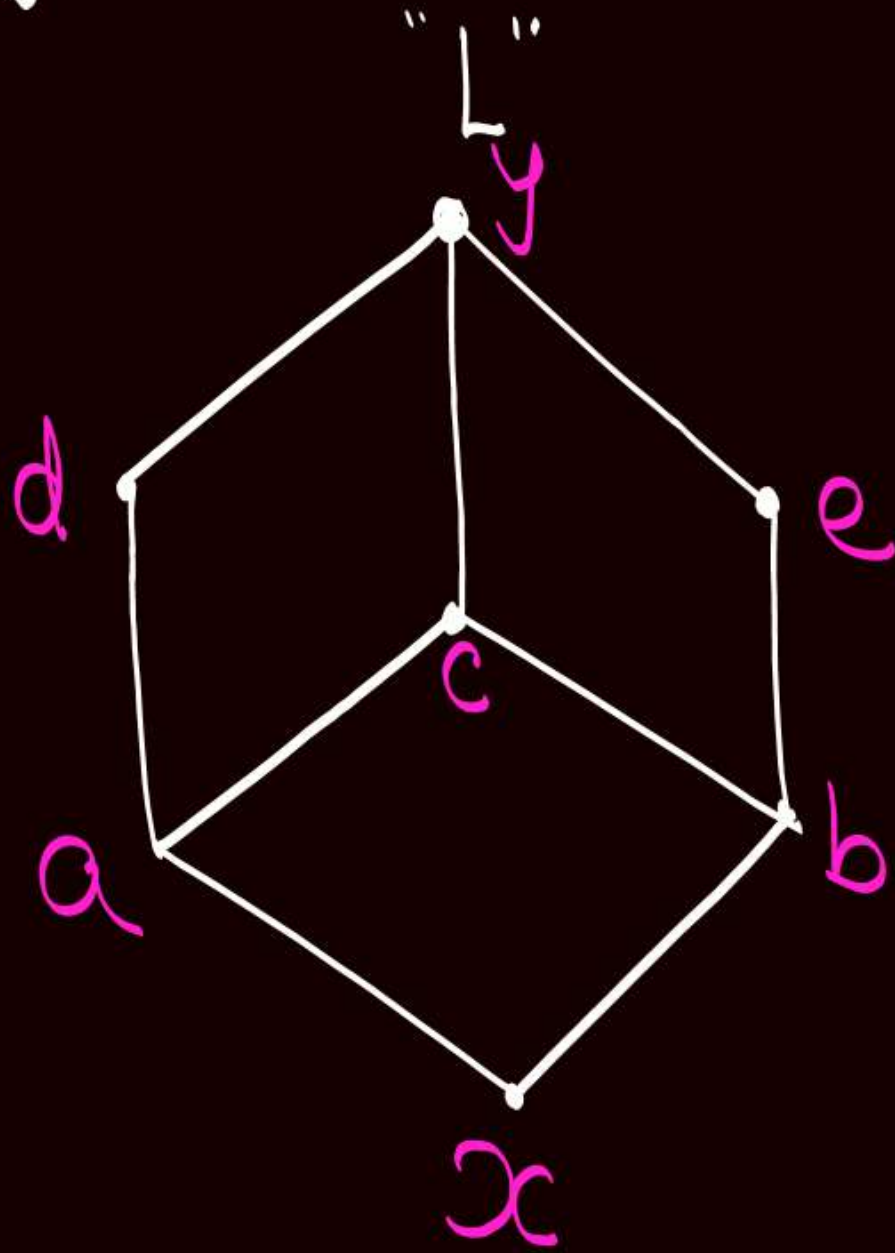
$$\text{glb}(a, e) = x$$

Same as L

It is a lattice

luc & glb is same as original lattice for every pair of elements
 $\therefore L_4$ is sublattice of L.

eg Consider the following Lattice 'L'



Which of the following is/are sublattices of lattice L

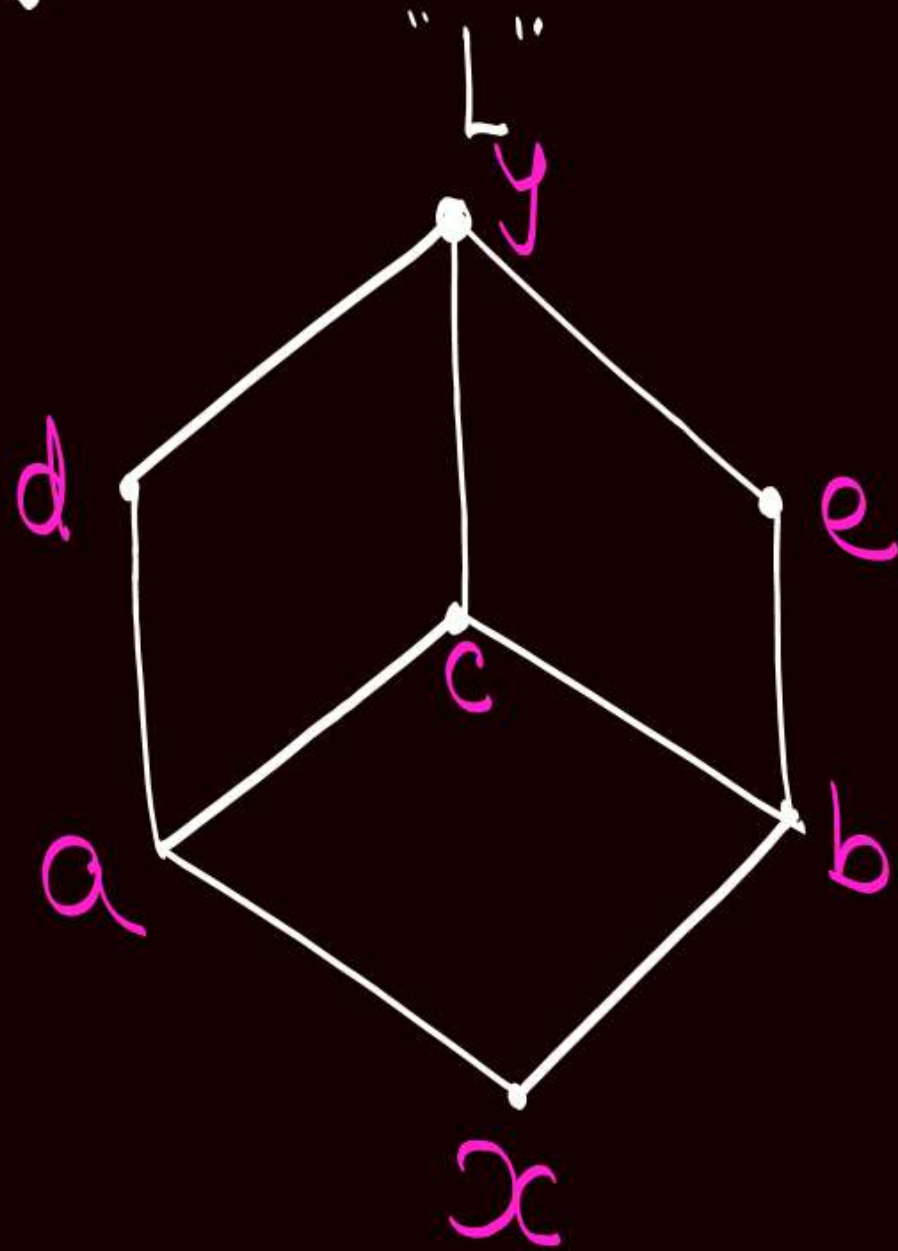
(i) $L_1 = \{x, a, b, y\}$

(ii) $L_2 = \{x, c, d, y\}$

☒ (iii) $L_3 = \{x, a, c, y\}$

☒ (iv) $L_4 = \{x, a, e, y\}$

eg Consider the following Lattice 'L'



Which of the following is are
sub-lattices of lattice L

(a) $L_1 = \{a, e, y\} =$

(b) $L_2 = \{a, b, c\} =$

(c) $L_3 = \{x, a, d\} =$

(d) $L_4 = \{a, c, y\} =$



2 mins Summary



✓
Topic

Complements of an element in a lattice

✓
Topic

Sub-lattice

THANK - YOU